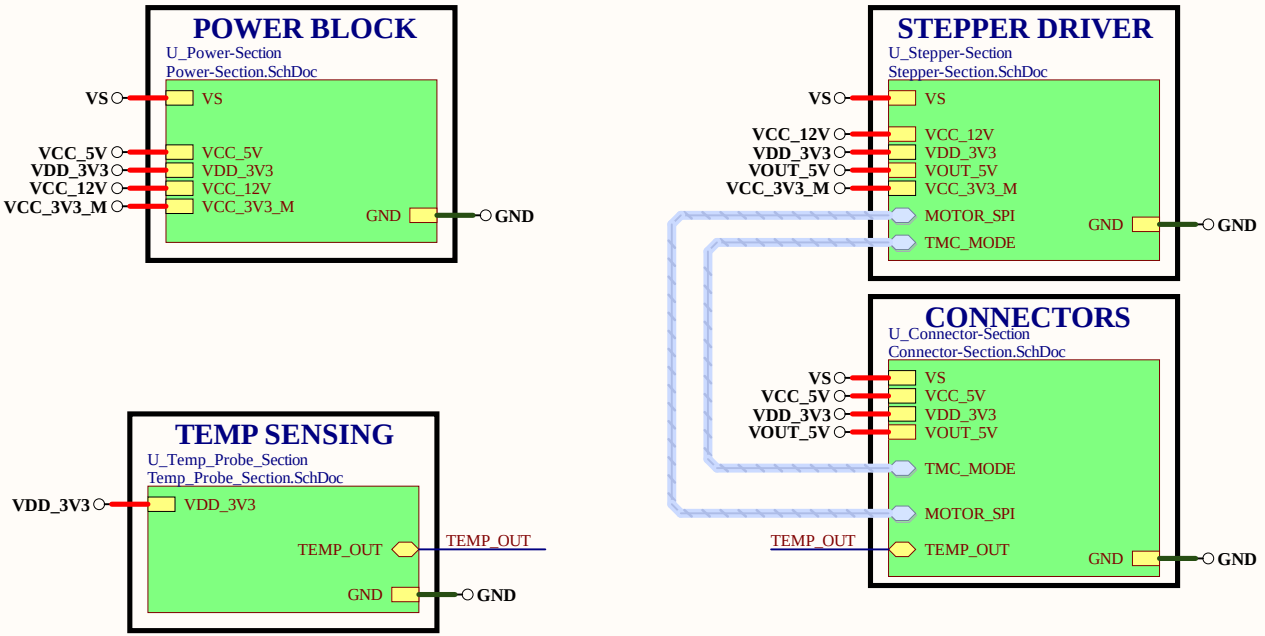
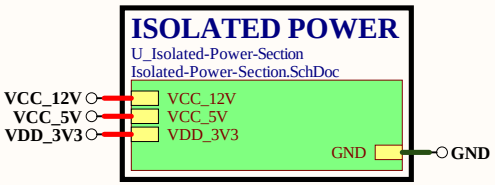
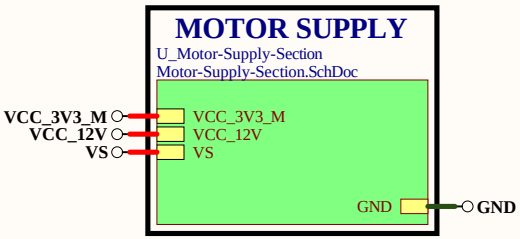
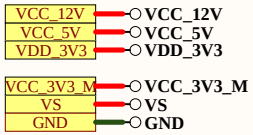
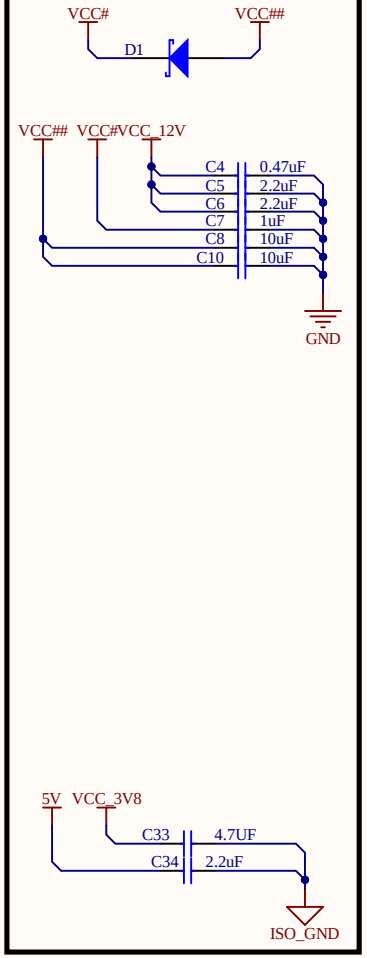

POWER BLOCK
U_Power-Section
Power-Section.SchDoc
VS
VCC_5V
VDD_3V3
VCC_12V
VCC_3V3_M
GND
GND
STEPPER DRIVER
U_Stepper-Section
Stepper-Section.SchDoc
VS
VCC_12V
VDD_3V3
VOUT_5V
VCC_3V3_M
MOTOR_SPI
TMC_MODE
GND
GND
CONNECTORS
U_Connector-Section
Connector-Section.SchDoc
VS
VCC_5V
VDD_3V3
VOUT_5V
TMC_MODE
MOTOR_SPI
TEMP_OUT
GND
GND
TEMP SENSING
U_Temp_Probe_Section
Temp_Probe_Section.SchDoc
VDD_3V3
TEMP_OUT
GND
GND
TEMP_OUT


ISOLATED POWER
U_Isolated-Power-Section
Isolated-Power-Section.SchDoc
VCC_12V
VCC_5V
VDD_3V3
GND
VCC_12V
VCC_5V
VDD_3V3
GND
GND

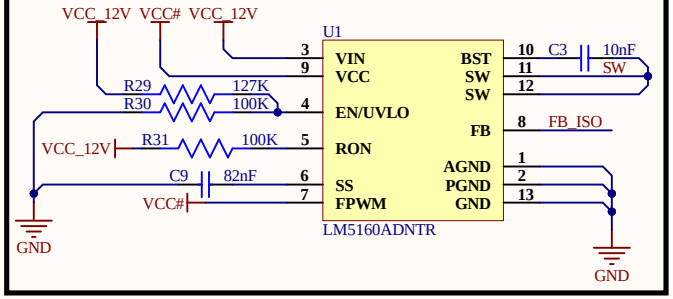

MOTOR SUPPLY
U_Motor-Supply-Section
Motor-Supply-Section.SchDoc
VCC_3V3_M
VCC_12V
VS
VCC_3V3_M
VCC_12V
VS
GND
GND


VCC_12V
VCC_5V
VDD_3V3
VCC_12V
VCC_5V
VDD_3V3
VCC_3V3_M
VS
GND
VCC_3V3_M
VS
GND

DECOUPLING CAPS

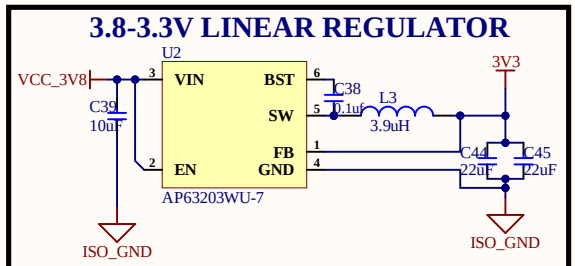


12V-3.8V ISOLATED BUCK CONVERTER

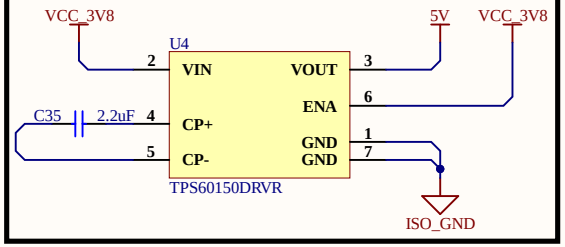


Primary side output voltage would be 5.7 for a 3.8V VSec (NP/Ns=Vpri/Vsec)
FB resistors RF2=3.7K and RF1=2K.
Max current of this transformer is 840mA. (suggested in datasheet)

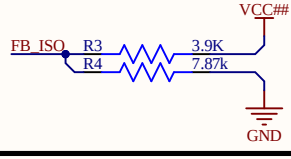
3.8-3.3V LINEAR REGULATOR



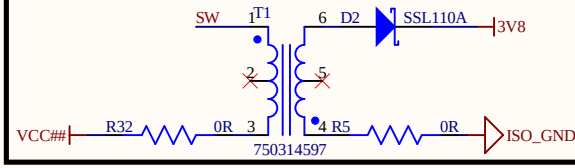
3.8-5V CHARGE PUMP



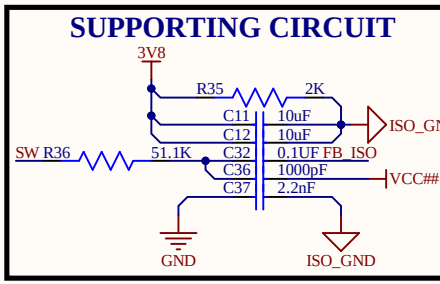
FEEDBACK NETWORK



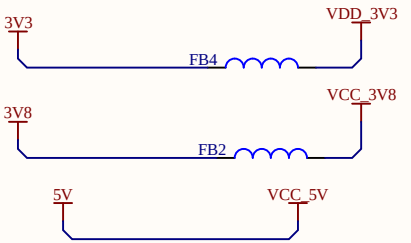
FLYBACK TRANSFORMER

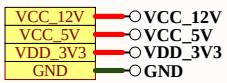


SUPPORTING CIRCUIT

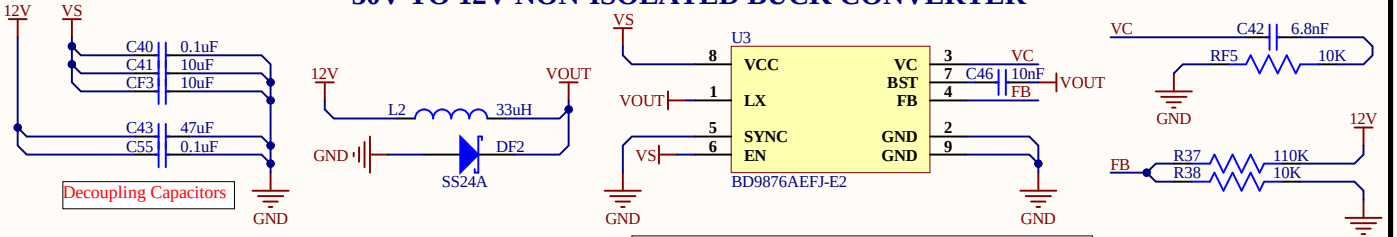


FERRITE CONNECTIONS

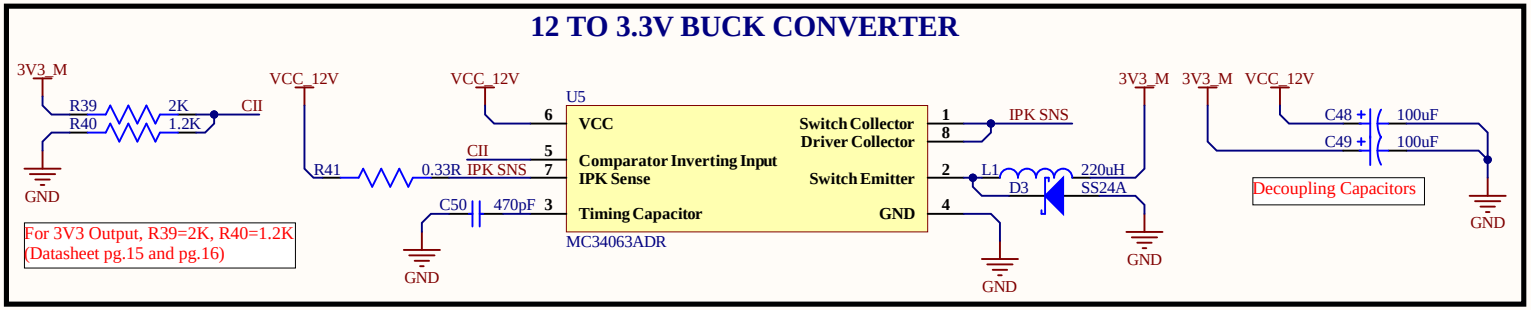




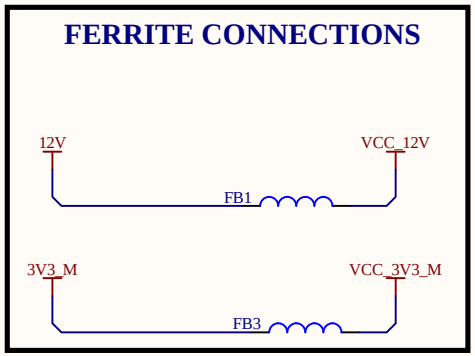
30V TO 12V NON-ISOLATED BUCK CONVERTER

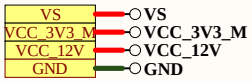


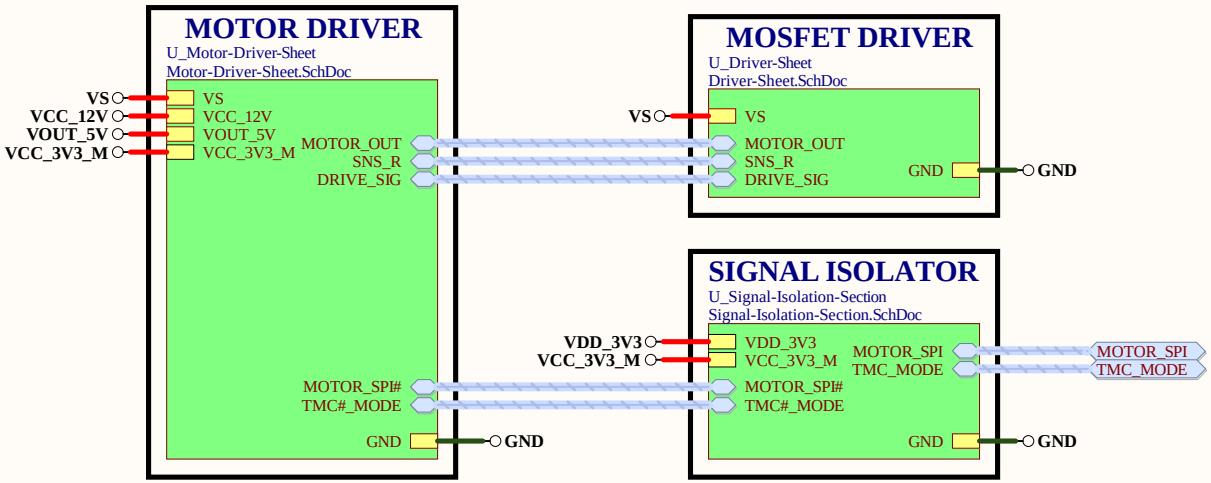
12 TO 3.3V BUCK CONVERTER

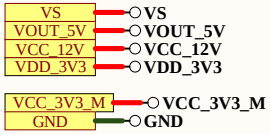


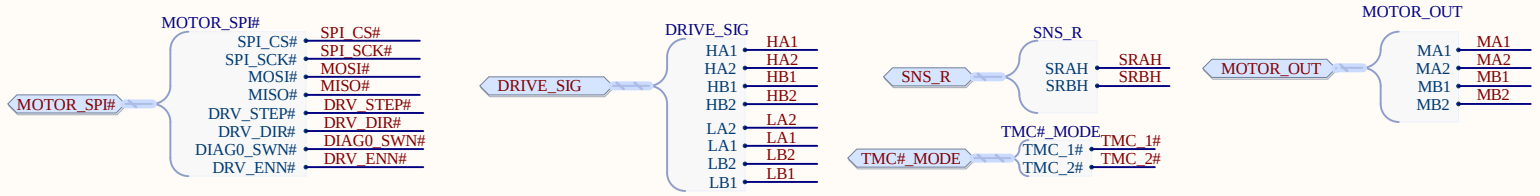
FERRITE CONNECTIONS



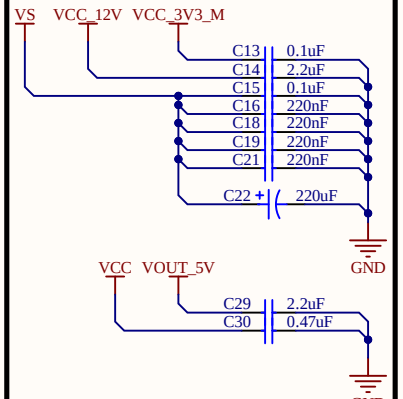



MOTOR DRIVER
U_Motor-Driver-Sheet
Motor-Driver-Sheet.SchDoc
VS
VCC_12V
VOUT_5V
VCC_3V3_M
MOTOR_OUT
SNS_R
DRIVE_SIG
MOTOR_SPI#
TMC#_MODE
GND
MOSFET DRIVER
U_Driver-Sheet
Driver-Sheet.SchDoc
SIGNAL ISOLATOR
U_Signal-Isolation-Section
Signal-Isolation-Section.SchDoc
VDD_3V3
MOTOR_SPI
TMC_MODE


VS
VOUT_5V
VCC_12V
VDD_3V3
VCC_3V3_M
GND

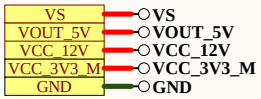


DECOUPLING CAPS

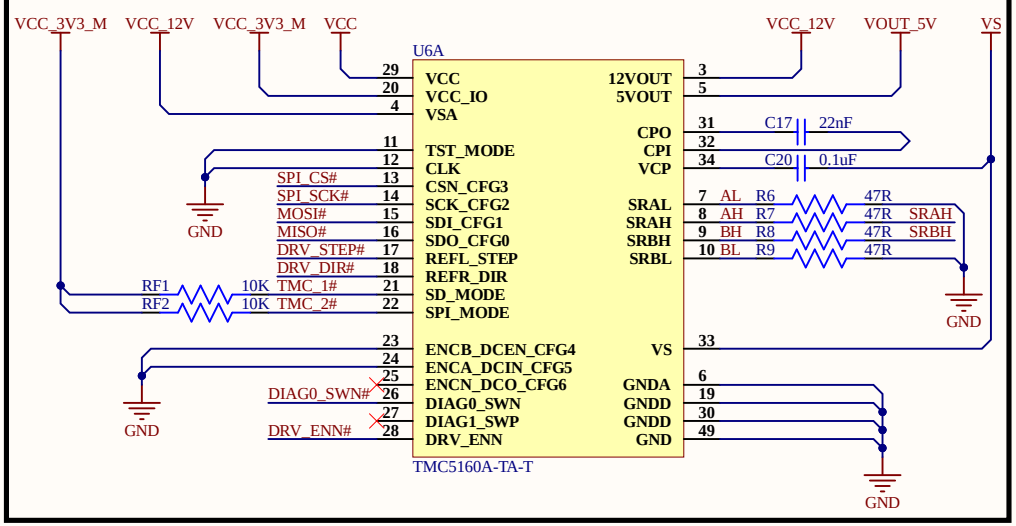


SERIES RESISTOR

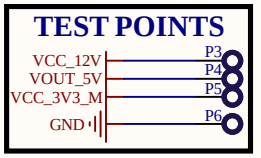




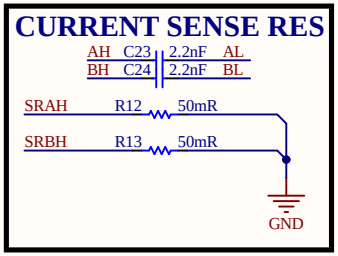
STEPPER MOTOR DRIVER-PART-A



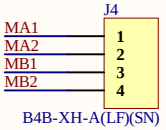
TEST POINTS



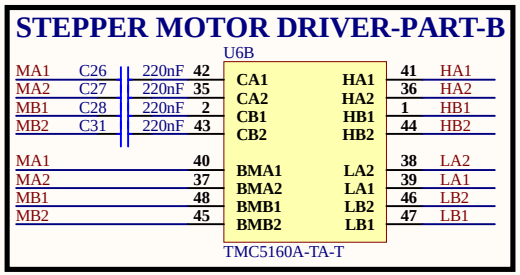
CURRENT SENSE RES



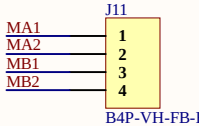
MOTOR OUTPUT

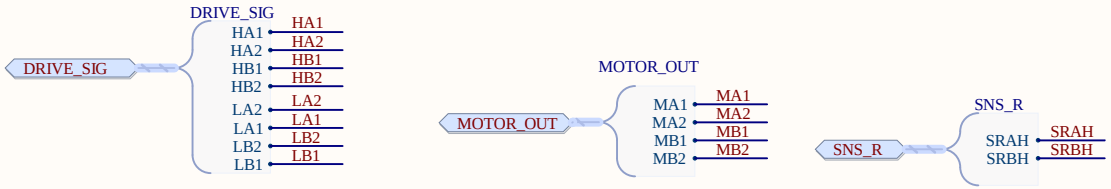


STEPPER MOTOR DRIVER-PART-B

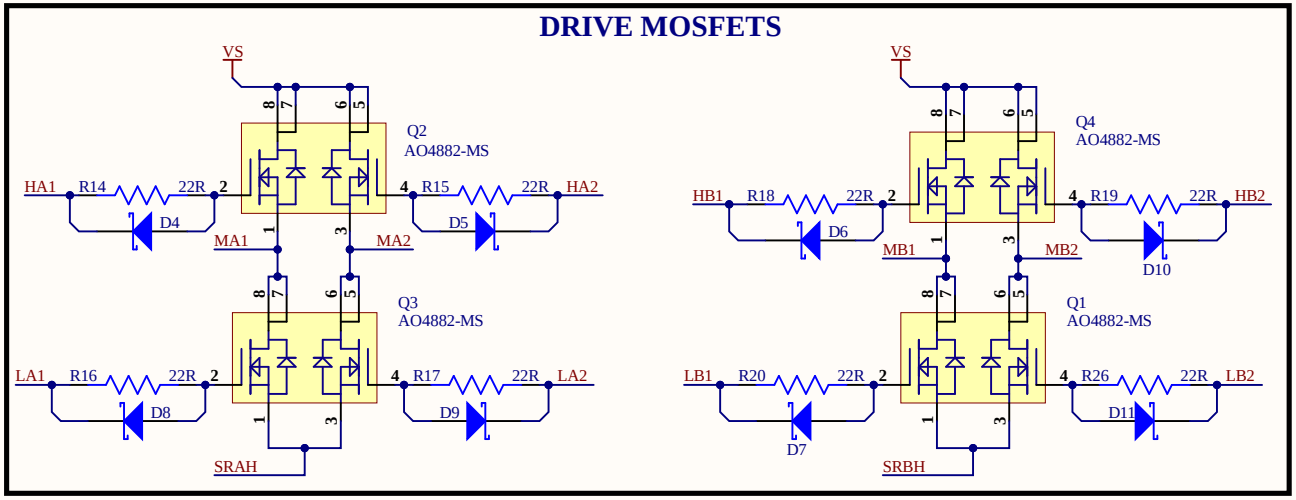


MOTOR OUTPUT FOR NEMA34

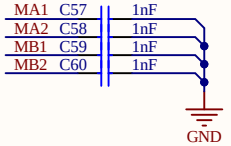




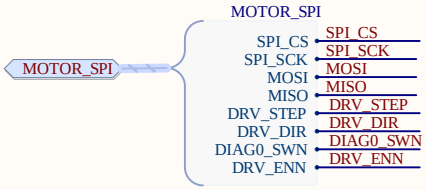
DRIVE MOSFETS

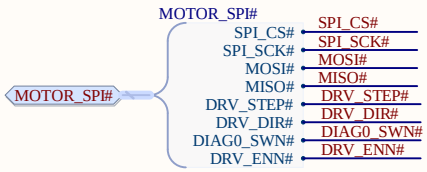


DECOUPLING CAPS



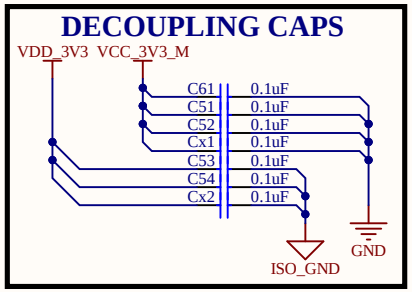


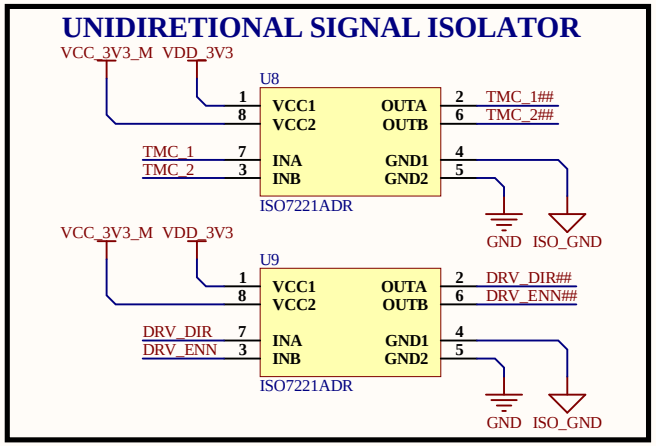


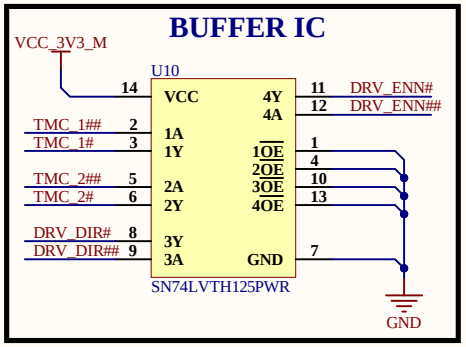


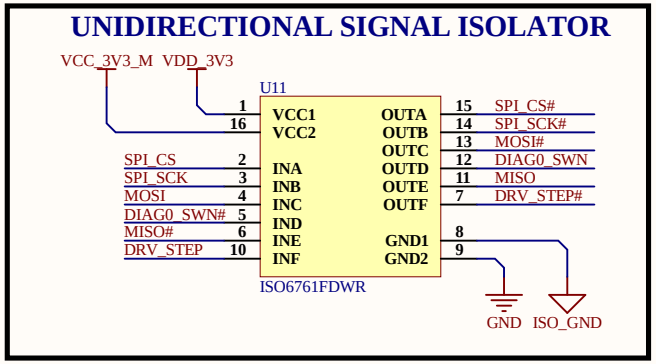










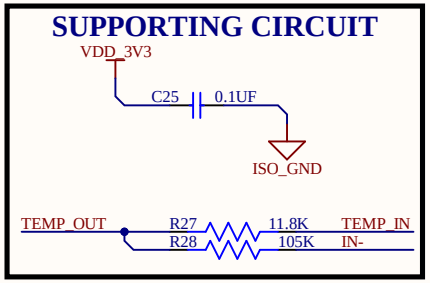




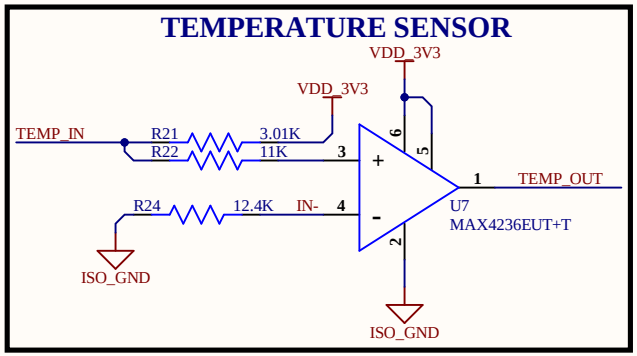
Title		
Size: A4	Number	Revision
Date: 11/28/2023		Sheet of
File: C:\Users\..\Signal-Isolation-Section.SchDoc		Drawn By:

TEMP_OUT TEMP_OUT

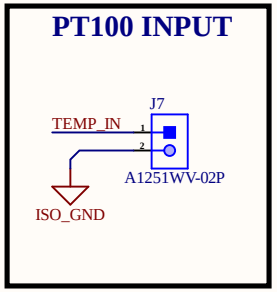
SUPPORTING CIRCUIT



TEMPERATURE SENSOR



PT100 INPUT



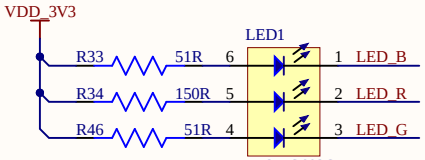
VDD_3V3 VDD_3V3
GND GND

INDICATION FOR MOTOR 5V SUPPLY

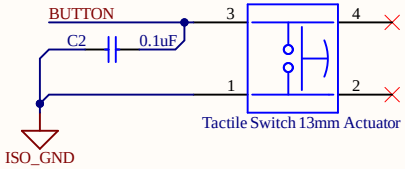
TMC_MODE
TMC_MODE
TMC_1
TMC_2
TMC_1
TMC_2
TEMP_OUT
TEMP_OUT

MOTOR_SPI
MOTOR_SPI
SPI_CS
SPI_SCK
MOSI
MISO
DRV_STEP
DRV_DIR
DIAG0_SWN
DRV_ENN
SPI_CS
SPI_SCK
MOSI
MISO
DRV_STEP
DRV_DIR
DIAG0_SWN
DRV_ENN

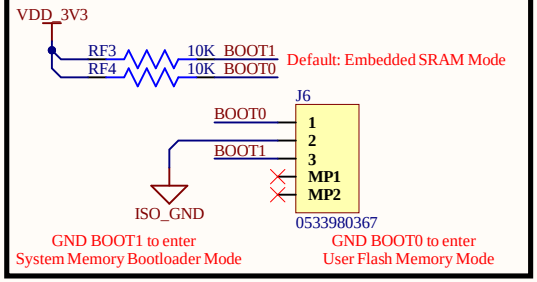
3.3V RGB LED



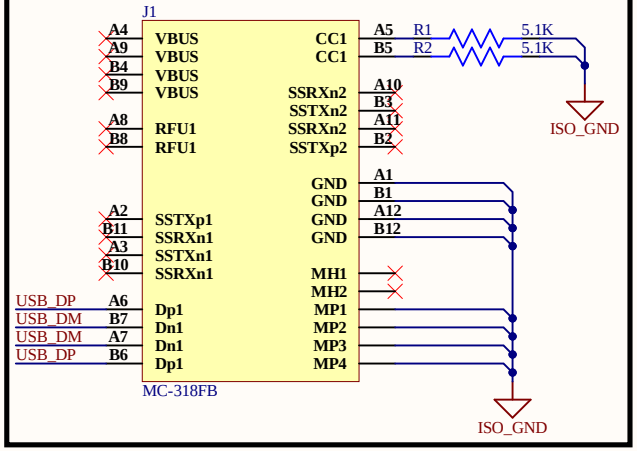
13 MM ACTUATOR



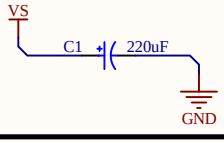
HEADERS TO SELECT BOOT MODE



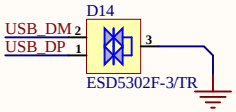
USB TYPE C FOR DATA TRANSFER



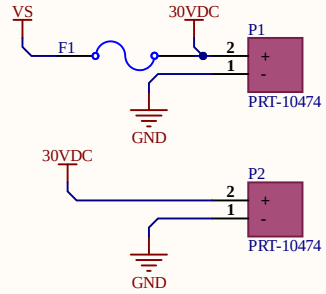
DECOUPLING CAP



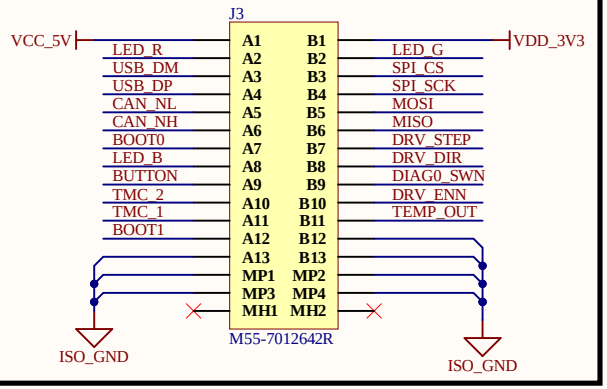
ESD PROTECTION



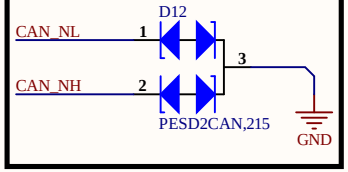
INPUT CONNECTORS



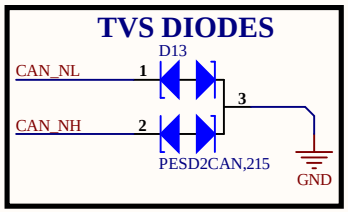
BOARD TO BOARD CONNECTOR



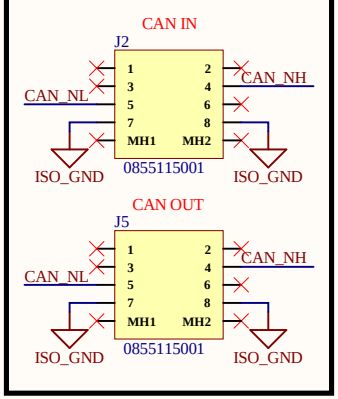
TVS DIODES



TVS DIODES



RJ45 FOR CAN BUS



VS
VCC_5V
VDD_3V3
VOUT_5V
GND
VS
VCC_5V
VDD_3V3
VOUT_5V
GND